

Kaelin Wulf

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Education

Reed College

Aug 2020 – May 2024

B.A., Neuroscience; Cumulative GPA: 3.51

Commendation for Excellence in Scholarship

Thesis: [Glia Cell Dynamics in the Zebrafish Optic Tectum](#)

Research Experience

Post-Baccalaureate Microscopy Engineer

July 2024 - present

Allen Institute for Neural Dynamics; Advisor: Adam Glaser

- Developed, characterized, aligned and operated custom large-scale light-sheet microscopes for imaging whole expanded mouse brains.
- Authored an imaging [protocol](#) and detailed documentation [wiki](#) on GitHub; and drafted a manuscript on the microscope and properties of expansion microscopy (in-progress).
- Improved optical and mechanical designs through CAD modeling, machining, and prototyping novel optical assemblies.
- Lead the assembly of multiple microscope systems and independently performed alignment and troubleshooting.
- Presented work at O-STEM, Optica Bio-Photonics Congress, Lake Conference, and SPIE Photonics West (planned Jan 2026).

Senior Thesis Student

August 2023 - May

2024

Reed College Neuroscience Department; Advisor: Kara Cervený

- Conducted a yearlong independent thesis research project studying the cell dynamics and patterns of microglia and oligodendrocyte progenitor cells (OPCs) in the optic tectum of larval zebrafish.
- Optimized an image acquisition and analysis pipeline to work with large data on limited hardware and trained team on best imaging practices.
- Produced a ~100-page written thesis with oral defense; presented results as a poster at the SDB, 2024 Northwest Regional Meeting, and as a selected talk to the Reed College community.

Undergraduate Research Fellow

May - Aug 2023

Reed College Neuroscience Department; Advisor: Kara Cervený

- Proposed, managed, and executed a summer research project on cell death and microglia cell dynamics in the optic tectum of larval zebrafish.
- Generated transgenic crosses and single-cell injectable plasmid vectors and imaged larvae using confocal and light-sheet microscopy.
- Presented findings as a poster at the Society for Neuroscience 2023 conference.

Undergraduate Research Fellow

May - Aug 2022

Reed College Biology Department; Advisors: Erik Zornik & Charlotte Barkan

- Led a collaborative research project investigating androgen-dependent neuroanatomy in *Xenopus laevis* vocal-motor circuits.
- Conducted animal surgeries, dissections, and histological preparations including IHC and tissue clearing for light-sheet imaging.

- Presented results with a poster at the Society for Neuroscience 2022 conference

Student Intern

Jun - Aug 2020

Air Force Research Laboratory; Advisor: Wellesley Pereira

- Collected and analyzed photometric telescope images of satellites to develop an approach to extracting fundamental physical properties of space objects.
- Collaborated with remote teams under virtual limitations to design novel methods for multispectral data analysis.

Additional Work Experience

Confidential Advocate and Coordinator

Sept 2022 – May 2024

Reed College SHARE program

- Provided confidential advocacy and support for survivors of interpersonal violence
- Facilitated weekly full team meetings, and led working groups focused on education and reshaping campus culture around topics of sexual violence
- Organized community events and led workshops on consent, supporting survivors, and the neurobiology of trauma
- Delivered workshops on survivor support, consent culture, and the neurobiology of trauma.

Music Technology Library Staff

Sept 2023 – May 2024

Reed College Performing Arts Resource Center

- Supported patrons in selecting, operating, and troubleshooting music and audio equipment
- Routinely operated and maintained shared studio space, applying skills in audio engineering and electronics.

Honors/Awards

Summer Undergraduate Research Fellowship 2022 & 2023

The Beckman Foundation, and the Helen Stafford Summer Research Fellowship Fund

Undergraduate Research Opportunity Grant 2022 & 2023

The Galakatos Science Research Fund, and Paul K. Richter & Evalyn Elizabeth Cook Richter Memorial Fund

Initiative Grant in Undergraduate Research 2024

Reed College, Neuroscience Department

Microscopy Image Competition 2024

Northwest Developmental Biology Conference, Nikon Exhibition

Publications

Wulf, K., Jiang, X., Glaser, A., et al. “An optimized expansion-assisted selective plane illumination microscope for nanoscale imaging of centimeter scale tissues”. (in preparation)

Conference Presentations and Proceedings

Wulf, K., Jiang, X., Woodard, M., Mwaniki, W., Glaser, A. “Optimized expansion-assisted selective plane illumination microscopy (ExA-SPIM)”. *SPIE Photonics West*, 2026. [Abstract & Talk]

Wulf, K., Javeri, R., Vasquez, S., Arshadi, C., Ouellette, N., Jiang, X., Baka, J., Kovacs, G., Woodard, M., Seshamani, S., Cao, K., Clack, N., Grim, A., Turschak, E., Liddell, A., Rohde, J., Grotz, P., Logsdon, M., Feng, D., Svoboda, K., Chandrashekar, J., and Glaser, A. “New technologies for mapping centimeter scale tissues with nanoscale resolution” *Lake Conference 2025*, [Poster].

Wulf, K., Jiang, X., and Glaser, A. “Characterization of a New Expansion-assisted Selective Plane Illumination Microscope”. *Optica Biophotonics Congress 2025*, Technical Digest Series (Optica Publishing Group, 2025), paper DM3A.2. [Abstract & Talk]

Jiang, X., **Wulf, K.**, Woodard, M., Mwaniki, W., Glaser, A. “A new expansion-assisted selective plane illumination microscope for nanoscale imaging of centimeter-scale tissues”. *Optica Biophotonics Congress 2025*, Technical Digest Series (Optica Publishing Group, 2025), paper NM5C.2.

Wulf, K., Jiang, X., and Glaser, A. “Expansion-assisted selective plane illumination microscopy for nanoscale imaging of centimeter-scale tissues”. *O-STEM*, 2024. [Abstract & Poster]

Wulf K. and Cervený K. “Effects of Innervation on Oligodendrocyte and Microglia Cell Dynamics in the Zebrafish Visual System”. *Society for Developmental Biology, Northwest Regional Meeting*, 2024. [Poster]

Wulf K., Yue J., and Cervený K. “Dynamics of Cell Death in the Denervated Zebrafish Optic Tectum”. Faculty for Undergraduate Neuroscience; *Society for Neuroscience*, 2023. [Poster]

Wulf, K., Truong, B.B., Zornik E., and Barkan C. “Examining Androgen Induced Changes to Vocal Neurons of Female *Xenopus Laevis*”. Faculty for Undergraduate Neuroscience; *Society for Neuroscience*, 2022. [Poster]

Professional Associations

Society for Neuroscience (SFN) 2022-present

Society of Photo-Optical Instrumentation Engineers (SPIE) 2025-present

Optica 2024-present

Skills

Engineering: Optical alignment, Basic CAD design, Soldering, Manual machining, Testing and characterizing optics, Specifying/purchasing Parts, Mechanical construction

Wet Lab: Animal surgery/dissection, PCR, Genotyping, Plasmid cloning, Microinjections, Tissue sectioning, Immunolabeling, Tissue clearing

Software: Python, R, Photoshop/Illustrator, 3D Image Analysis (Imaris), ImageJ/FIJI